



Enriching BIM models with fire safety equipment using keypoint-based symbol detection in escape plans

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ABSTRACT

In the context of fire safety inspections, Building Information Modeling (BIM) models enriched with Fire Safety Equipment (FSE) components can be used to complete compliance checks and other analyses. However, BIM models often lack the required FSE information. To address this issue, escape plans are a convenient source of data, as they show the position and type of FSE on floor plans. Therefore, this study proposes an automated method to analyze escape plans and extract FSE component information to enrich existing BIM models. The method employs the deep learning model Keypoint R-CNN for symbol detection. Symbol locations are then translated into physical positions within the BIM model. Through a real-building case study, the method demonstrates promising results. Future research may focus on improving the symbol detection performance and the registration between the BIM models and fire escape plans, as well as utilizing the extracted information for actual fire safety analyses.

1. Introduction

The utilization of Building Information Modeling (BIM) [1,2] has gained significant traction in the design and construction phases of buildings, primarily attributed to its capacity to improve operational efficiency and information traceability, as well as support informed decision-making [3]. However, there is a growing recognition of the potential benefits that BIM offers in the Operations and Maintenance (O&M) phase as well [4], which encompasses activities like facility management [2,5–13], maintenance planning, and safety inspections [14–16]. In particular, regular fire safety inspections are vital for ensuring compliance with safety regulations and identifying potential hazards in buildings.

BIM can greatly facilitate these inspections by providing a detailed and interactive virtual representation of the building, enabling inspectors to navigate through its components, assess fire safety measures, and document the inspection results [15,17,18]. Also, the visualization of building interiors with Fire Safety Equipment (FSE) components comes in handy for the development of disaster prevention and mitigation systems that help in emergency decision-making [15,19]. As pointed out by previous studies [15,17,20–22], equipping augmented reality applications with building information bears great potential and could be a used for those visualization tasks.

Unfortunately, existing BIM models often lack accurate and up-to-date information about FSE. Therefore, they need to be enriched with

details on (at least) the positions and types of FSE, in order to be leveraged for fire safety inspections and analyses [23,24].

The retrospective creation and enrichment of BIM models for existing buildings are the subjects of intensive research efforts [25,26]. Among others, a popular strategy is to extract building information from 2D drawings. For instance, prior studies have addressed reconstructing building geometry from floor plans and other construction drawings [27–35]. Similarly, substantial efforts have been devoted to the detection and recognition of symbols in such drawings [36–43]. Therefore, as they have demonstrated effectiveness in related studies, it is reasonable to consider employing similar methods in the context of integrating FSE information into pre-existing BIM models.

Given their explicit purpose of assisting occupants during emergencies, escape plans emerge as a prime candidate for integrating FSE information into established BIM models. These plans inherently contain precise positioning and classification of FSE throughout a floor of a building. Conveniently, these plans adhere to standardized drawing conventions, making it considerably easier to identify and interpret the associated symbols [44]. Hence, it becomes possible to extract FSE information from escape plans, as demonstrated in [45], enabling their integration into the existing BIM model.

In this paper, we propose an approach to tackle the integration of FSE information into existing BIM models. Our approach utilizes

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advanced computer vision techniques to automatically extract the positions and types of FSE from escape plans. In this regard, keypoint-based deep learning models are used to accurately detect symbols in escape plans. In addition to merely locating symbol icons with bounding boxes, these models are also able to recognize the precise location as a single point on the drawing, as it may be indicated by auxiliary lines. By this enhanced inference output, the proposed approach overcomes a crucial limitation of prior approaches based on bounding box detection of FSE components [45]. Additionally, the textual information on the escape plan is extracted and used to determine the depicted building section to ensure the correct localization of the FSE. Subsequently, this information is integrated into the BIM model, enriching it with the extracted information.

The objective of this paper is to present the proposed approach, evaluate its effectiveness and accuracy through a case study, and discuss the implications and potential benefits of incorporating FSE information into BIM-based workflows. This research notably contributes to the ongoing efforts in as-is BIM creation by

- exploring the utilization of escape plans for FSE information extraction,
- proposing an end-to-end BIM enrichment method based on escape plans, and
- demonstrating the advantages of keypoint-based symbol detection in floor plans.

In line with related research efforts [26], this study has the potential to decrease manual labor and time required for updating BIM models with FSE equipment, consequently enhancing efficiency in as-is BIM modeling [46], and ultimately promoting BIM-based fire safety management [15,47].

The remainder of the paper is organized as follows: Section 2 introduces key fire safety concepts, gives an overview of prior works towards symbol detection in 2D drawings, and highlights the identified research gaps. The proposed methodology is presented in Section 3, combining methods for detecting the FSE in drawings and enriching the BIM model with the extracted 2D information. Section 4 shows the results of the presented methods by means of a case study, and evaluates the effectiveness and accuracy of the employed deep learning models. Implications, limitations, and future research directions are discussed in Section 5. Finally, Section 6 presents the paper's concluding remarks.

2. Background

2.1. Fire safety and inspections

Fire safety is a critical aspect of building management, encompassing a comprehensive set of measures aimed at preventing and controlling fires, ensuring the safety of individuals, and minimizing property damage [48]. To achieve these protection objectives, fire safety regulations are a cornerstone of national and regional building codes. These regulations dictate mandatory measures concerning building design, construction, and operation. While specifics may vary between countries, common elements include design requirements for accessible escape routes, structural measures to contain fire development, and proactive measures for defensive fire protection [49–53]. To meet these requirements, fire protection considerations must be integrated throughout all stages of a building's life cycle. Particularly in the O&M phase, ensuring continuous compliance with the fire protection objectives is crucial for both preventive and defensive fire protection. Therefore, conducting regular fire safety inspections is a legal responsibility to ensure the safety of occupants. Fire safety regulations mandate building operators to conduct regular inspections of fire safety-related systems and equipment, which involves physically visiting the facility and systematically assessing relevant assets to verify compliance. In general, neglecting these inspections can lead to increased safety hazards [54]. A valuable tool for this purpose is

a fire safety inspection checklist, aiding in the systematic evaluation of essential elements of structural, plant, organizational, and defensive fire protection in an existing building. While no standardized procedure covers all specific items, various guides and resources are available as working aids [17,21].

BIM models with FSE information can enhance the efficiency and efficacy of inspections. They do not just provide any existing inventory of FSE information, but can also help inspectors (and even robots) navigate to all components [55–58]. They can also serve a user-friendly interface to update any FSE information [59].

2.2. Fire safety-related Scan-to-BIM

Building upon the importance of fire safety inspections, integrating fire safety-related information into BIM models is a crucial aspect of advancing the O&M phase of a building. Various sensor-based and data-driven approaches are advocated in the literature for an efficient Scan-to-BIM process [60–64].

One notable approach by Vandecasteele et al. [61] utilizes video object recognition linked with BIM to enhance fire safety management, providing feedback on location estimation, fire location, evacuation guidance, and fire forecasting. Moreover, researchers have specifically investigated the recognition and pose estimation of Mechanical, Electrical, and Plumbing (MEP) components in 3D point clouds [60,62,65]. Notably, Adan et al. [62] highlight the limitations of using orthoimages generated from point clouds for component detection, emphasizing the need for alternative strategies. In other studies [66,67], researchers focus on training object detection neural networks to identify fire assets like extinguishers and exit signs in images, leveraging transfer learning from the COCO dataset weights.

As the above review shows, despite a growing interest in automating the Scan-to-BIM process for MEP and FSE, existing literature predominantly focuses on on-site data acquisition techniques like interior photography or LiDAR scanning. However, a compelling argument exists for leveraging building documentation instead, given its ease of accessibility and inclusion of valuable semantic information. Consequently, there is merit in exploring whether the application of 2D plan analysis, a technique successfully employed in tasks such as symbol detection in floor plans or other technical drawings [32,34–43,68], could benefit Scan-to-BIM in this context. While Hassaan et al. [45] proposed a digitization method for escape plans, including detection of FSE symbols, key limitations require further exploration. For example, if an FSE component's location is indicated by an auxiliary line, it cannot be inferred by existing methods. Furthermore, additional information on the escape plans, e.g., building story numbers, are not considered. Thus, this study investigates the subject further, addressing the identified research gaps.

2.3. Escape plans

Escape plans (see example in Fig. 1) play a crucial role in informing individuals within a building about available escape routes, thus ensuring the safety of occupants in the event of a fire. In accordance with the guidelines set by ISO 23601 [44], these plans are specifically designed for public areas and workplaces. While they may not delve into the intricacies of detailed technical drawings meant for specialized professionals, they provide clear and concise information about escape routes, emergency exits, and the locations of fire safety equipment.

To maintain a clear and consistent representation of equipment in plans, international and national requirements adhere to standards for safety colors and signs, as outlined in ISO 7010 [69]. Thus, if first aid and fire-fighting facilities are available in the building, the plan should utilize safety signs that comply with ISO 7010. Fig. 1 showcases the application of standardized symbols and color codes for clarity. This standardized representation proves advantageous for deep learning-based symbol detection techniques, as it facilitates a low generalization error. Also, since escape plans are required by law to be readily available and displayed in strategic locations in buildings, they are a convenient source of FSE information for this study.



Fig. 1. Example escape plan.

2.4. Research gaps and formulation of research problems

The preceding sections highlight the significance of integrating fire safety information into BIM models, particularly focusing on the utilization of escape plans as a potential data source. However, several research gaps and challenges have emerged. First, while there is a trend towards automating Scan-to-BIM for fire safety and MEP components, current methods rely on on-site data gathering, such as photography or LiDAR scanning. This raises questions about the viability of using alternative sources like escape plans, which may offer valuable semantic information and easier accessibility. Second, existing digitization methods for escape plans have limitations. For instance, they struggle with auxiliary lines indicating FSE locations and overlook additional information like building story numbers. To address these issues, we pose the following research questions:

- How can escape plans be used to automatically integrate fire safety data into BIM models?
- What are the limitations of current escape plan digitization methods, and how can they be improved for accurate FSE extraction?
- How can advanced techniques like keypoint-based object detection enhance FSE extraction from escape plans, considering standardized symbols and color codes?

Addressing these research problems aims to contribute to the development of more efficient methods for enriching BIM models with fire safety-related information, thereby enhancing building management practices.

3. Methodology

An overview of the proposed process to incorporate FSE information into BIM models is depicted in Fig. 2. First, an escape plan is captured by the user (A) and pre-processed for further analysis (B). Details on these steps are provided in Section 3.1 and Section 3.2, respectively. Next, the required information is extracted from the plan. First, Section 3.3 introduces a keypoint-based deep learning symbol detection network for localizing the FSE on the plan (C). Then, Section 3.4 details how the text information on the plan is utilized to infer the depicted building floor (E). To obtain the FSE components' model coordinates, the captured image is registered with the geometric model of the building floor (D), and the corresponding coordinate transformation is applied to the pixel coordinates of the located symbol positions (F). Finally, the enrichment of the BIM model with the FSE components is described in Section 3.6 (G). Each step of the pipeline is implemented in the course of this study, and the test results and a case study demonstration are given in Section 4.

3.1. Escape plan capturing

The proposed method prioritizes simplicity in application, aiming to provide a user-friendly solution. Therefore, it does not rely on meticulously curated high-quality scans of escape plans. Instead, a basic smartphone image of the wall-mounted escape plans is sufficient, as long as the resolution enables precise positioning of FSE. The assumption is made that each input image captures the entire wall-mounted escape plan without significant rotation, distortion, or lighting reflections.

3.2. Region cropping

Following the capture of escape plans, the next step is to prepare the image for FSE symbol detection. In this regard, it is necessary to crop the image to the actual floor plan including the FSE locations, since the escape plan also includes additional text and a legend for all symbols, which could result in false positives during equipment localization. Therefore, the image undergoes a deep learning-based Region of Interest (ROI) cropping step, focusing on the actual floor plan. In this study, the ROI is defined as the area enclosed by the smallest axis-aligned rectangular bounding box of the escape plan, which encompasses the entirety of the floor plan geometry. Typically, this excludes significant portions of the legend, plan header, and other supplementary text elements. An example ROI, derived from the example escape plan in Fig. 1, is shown in Figure Fig. 3.

Previous studies have already shown the successful application of YOLO models for engineering drawing view separation problems [70,71]. In this study, a YOLOv8¹ object detection model is employed to automate the ROI cropping step. The model is trained to identify and delineate the floor plan regions on escape plans, enabling the extraction of these regions from the image for further processing.

3.3. FSE symbol detection

The next crucial task is the precise localization of FSE symbols within the escape plan. Certain drawing characteristics suggest the use of keypoint-based detection methods, as illustrated in Fig. 4. In some instances, the symbol's location on the plan aligns with the corresponding physical FSE component's location in the building, as demonstrated in Fig. 4(a). However, there are cases where the symbol icon is not directly placed in the drawing corresponding to the physical component location. Instead, it may be situated nearby, with an auxiliary line indicating the exact location (true location) [45], as shown in Fig. 4(b). When multiple FSE components share the same location, they are grouped by an encapsulating box, and their position is denoted by a single shared auxiliary line (cf. Fig. 4(c)). Finally, some symbols represent multiple instances of the component and, therefore, indicate additional true locations as seen in Fig. 4(d).

Given these nuances, when a mismatch occurs between the FSE component's actual location and the symbol icon, an additional point must be detected, which calls for a keypoint-based detection approach rather than bounding box-based detection. Moreover, the manual capturing of the plans imposes slight distortions on the drawing, leading to skewed symbol icons, which cannot be mimicked with axis-aligned bounding boxes. Thus, bounding box-based labeling does not suffice to specify the symbol location in all cases. Instead, we propose the use of a keypoint-based annotation scheme, a methodology that has shown promising results in similar studies [43].

Detailed descriptions of the developed symbol detection method are provided in the following sections. First, Section 3.3.1 specifies the annotation scheme utilized to create training data for the keypoint detection network based on the acquired real escape plan imagery

¹ <https://github.com/ultralytics/ultralytics> (Accessed: 15 December 2023)

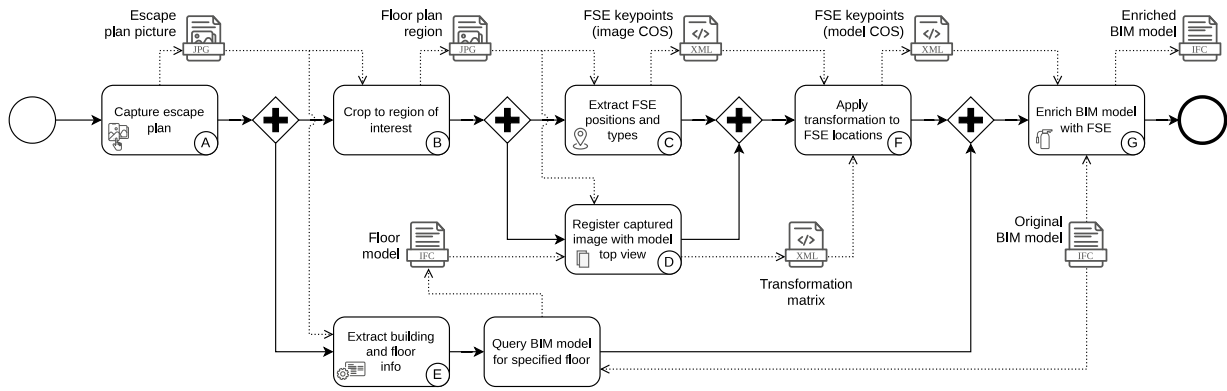


Fig. 2. Overview of the proposed pipeline.

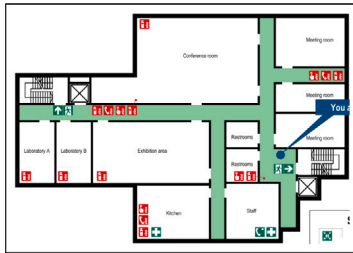


Fig. 3. Example ROI of an escape plan as defined in this study.

3.3.1. Labeling strategy

In the context of FSE symbol detection, we draw inspiration from the human pose estimation task to establish a labeling strategy, as illustrated in Fig. 5. For each symbol, a skeleton of 5 keypoints is created. Keypoints 1–4 represent the four corners of the symbol icon, while keypoint 5 defines the position indicated by the position marker on the escape plan. Special attention is given to two cases:

- When an FSE symbol has two position markers, denoting two components of the same kind at different locations, two skeleton instances are created. Keypoints 1–4 of both instances share the same locations in the image, ensuring the capture of all instances, even if only one symbol is depicted.
- If no position marker is present for an FSE symbol, the position of the component is assumed to be the closest wall. In such cases, keypoint 5 is set accordingly to represent this position.

This approach addresses potential ambiguities and ensures the comprehensive capture of all instances of FSE symbols. In instances where no position marker is available, the proposed method provides a reliable approximation of the component's location, similar to what human perception might yield in such scenarios.

3.3.2. Training data acquisition

The next step is the acquisition of training data to train the symbol detection model. For this purpose, we collected images from escape plans in the campus buildings of the Ruhr University Bochum. To ensure the relevance and effectiveness of the training data, we only included escape plans adhering to the current symbol standard defined in ISO 23601 [44]. This dataset, referred to as the *campus images* dataset, comprises 167 distinct plans. After annotation, these plans exhibit 2,398 labeled FSE symbols. The class distribution is presented in Table 1.

3.3.3. Synthetic data generation

In an effort to increase the size and diversity of the training set, a total of 108 web images of escape plans are processed such that they appear as additional training images. Fortunately, owing to the standardized design of escape plans, these web images exhibit a similar style to the acquired campus images. However, as most of them are clean, digital plans, certain transformations are applied to align them with the appearance of the training images.

Assuming an example image as input for the transformations, the entire synthetic training image creation process is illustrated in Fig. 6. First, the input image is modified by adding Gaussian noise. Also, a background template is randomly selected from a prepared template set. Each template has an image layer including a placeholder for the escape plan and the surrounding image, usually a building interior wall. Second, the dimensions and shape of the placeholder are used to apply a perspective transformation to the input escape plan. Third, a lighting

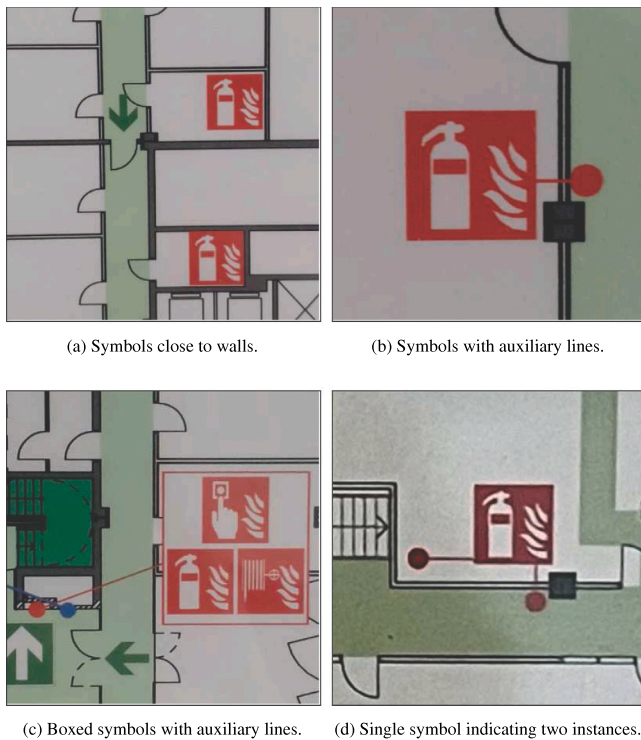


Fig. 4. Exemplary FSE symbols with different location specifications.

(Section 3.3.2). Subsequently, Section 3.3.3 introduces a tool for generating synthetic training images alongside real training images. The deep learning model used for keypoint detection is briefly introduced in Section 3.3.4. Finally, Section 3.3.5 documents the training setup of the detection model.

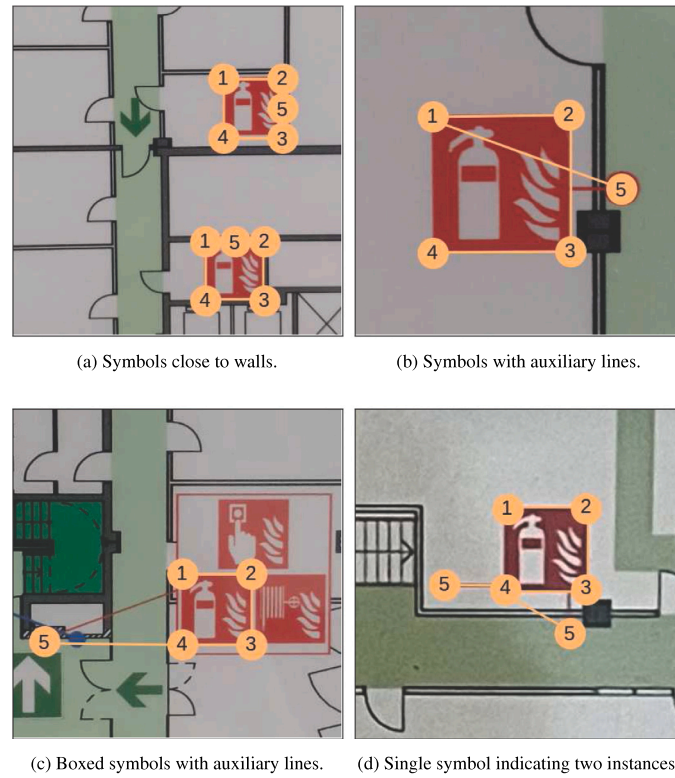


Fig. 5. Annotation schemes for FSE symbols with different location specifications. Compare with Fig. 4 for original images.

Table 1
FSE symbol class distribution in collected dataset.

Symbol	Name	# In collected images	# In synthetic images	# Total	Frequency
	Fire extinguisher	1,573	692	2,265	64,75%
	Manual call point	644	324	968	27.24%
	Fire hose reel	181	72	253	7.12%
	Fire fighting equipment	0	41	41	1.15%
	Emergency phone	0	25	25	0.70%
	Fire ladder	0	1	1	0.03%

effect is added to the escape plan. All three effects are supposed to enhance the realism of the web image. Finally, the modified escape plan is seamlessly integrated onto the selected background template, resulting in the final training image. This augmentation process contributes to the increased diversity and robustness of our training set.

3.3.4. Keypoint detection model

As a deep learning architecture for keypoint detection, this study employs the Keypoint R-CNN, a variant of Mask R-CNN [72]. The structure of the model is visualized in Fig. 7. In addressing the specific challenges of localizing and classifying FSE symbols within escape plans, the Keypoint R-CNN stands out as a robust solution. Leveraging

the architecture of Mask R-CNN, this model is tailored to precisely detect objects in images by means of skeletons with n keypoints. In this study, $n = 5$ keypoints is chosen to account for the 4 corner points of FSE symbols plus the location of the component, allowing for a complete extraction of FSE related information from the plans.

This choice of model architecture has three main advantages. First, it enables the precise detection of non-axis-aligned bounding boxes, which occur as a result of perspective distortions arising with photographic capturing. Second, the multimodal loss function leads to improved performance, in comparison with a purely bounding box-based detection. Finally, it enables detection of position markers in addition to the symbols themselves.

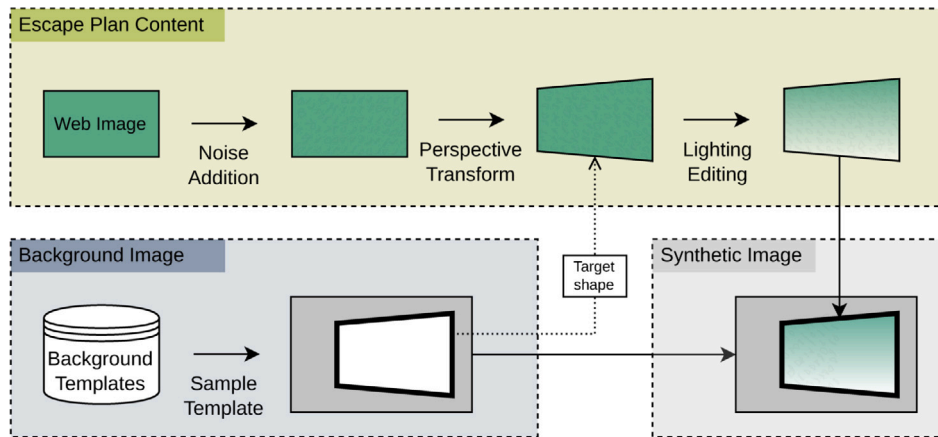


Fig. 6. Overview of the used synthetic image creation pipeline.

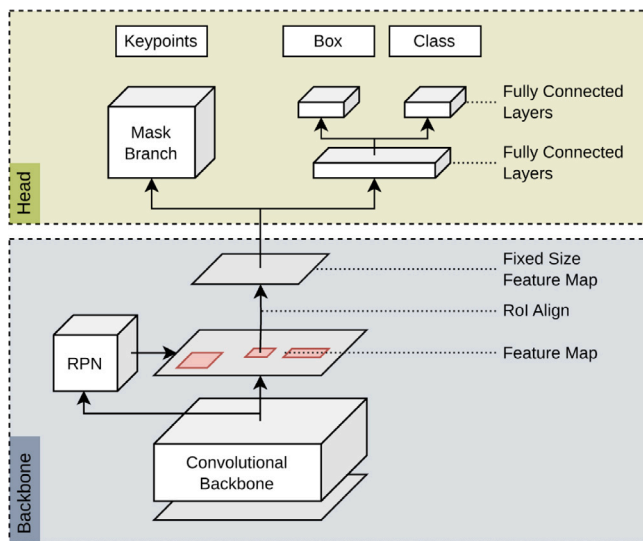


Fig. 7. Keypoint R-CNN detection model architecture (cf. [72]).

3.3.5. Training setup

Building upon the Keypoint R-CNN architecture described in Section 3.3.4, the model is initialized with ResNet50 [73] backbone pre-trained on the ImageNet dataset. The training dataset consists of campus and web images combined, while the validation set consists of 16 images from the campus images. During training, various data augmentation techniques, such as rotation (up to 5 degrees), blurring, random scaling, and brightness changes, are applied to enrich the training data. The model is evaluated by means of a separate set of campus images (cf. Fig. 8).

The training parameters are selected as follows: As an optimizer, the Nesterov Accelerated Gradient Descent (NAG) optimizer with a momentum of 0.9 is employed. The default learning rate of 0.0025 is maintained for 30 epochs. Additionally, weight decay regularization with a coefficient of 0.0001 is applied to prevent overfitting. All computations are conducted on an NVIDIA Tesla A100-SXM4-40 GB GPU.

3.4. Text recognition

The text information on the escape plan is extracted using the text detection and recognition pipeline proposed in [74], as it is found suitable for application on floor plans. This pipeline comprises a YOLOv7

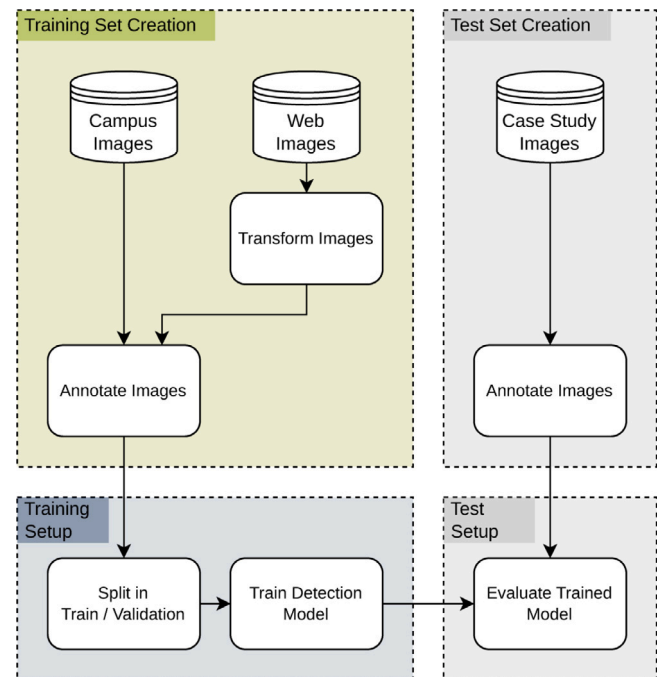


Fig. 8. Usage of datasets for training the Keypoint R-CNN detection model.

[75] text detection model and the Optical Character Recognition (OCR) engine PARSeq [76].

By applying text detection and recognition to the captured floor plan, a list of text elements on the drawing is obtained. This study specifically focuses on extracting information related to the current position. Notably, each escape plan typically includes a specification of the current building section and floor.

To precisely extract the required information and distinguish it from other text elements such as location names, legends, and instruction texts, regular expressions are searched. Regular expressions indicating floor information are utilized in the queries, such as: `'Floor\s?F?(\d+)'`, `'Level\s?(\d+)'`, and `'([0-9]+)(?:st|nd|rd|th) floor'`.

Since individual building complexes may feature unique naming and numbering schemes, these regular expressions should be configured for each building complex individually. Furthermore, the floor descriptors must align with the floor names in the BIM model, stored in Industry Foundation Classes (IFC) format, to establish automatic linkage. With the extracted floor number, the building model can now

be queried for the corresponding building story to be enriched with FSE information.

3.5. Image registration

Before incorporating FSE components into the existing building model, precise knowledge of their locations in the Coordinate System (COS) of the BIM model is essential. However, these component locations are initially known only with respect to the image COS after the symbol detection step. Therefore, a transformation is necessary to map these locations from the image COS to the model COS.

Assuming that the escape plan accurately aligns with the outlines of the BIM model, the following steps are necessary:

- Generating a plan view of the building story from the BIM model, similar in style to the escape plan.
- Registering the escape plan image to the generated plan view.
- Defining a valid transformation from the pixel COS in the generated top view image to the model COS.
- Computing the FSE components' locations with respect to the model COS based on the previously defined transformation.

It is important to note that automating this part of the proposed pipeline is not within the scope of this study, and, instead, a method based on user interaction is employed. It is noted, though, that prior studies have successfully employed automated registration techniques in related contexts. For instance, Mahmood et al. [77] addressed the registration of as-planned and as-is floor layouts via point cloud segments; Trzeciak and Borrmann [78] proposed a drawing-to-model registration system based on geometric hashing, and Bueno et al. [79] achieved plane patch-based building model registration techniques in 3D space, a method which could possibly be adapted for 2D space. However, the following issues might impose difficulties in this particular registration task:

- The escape plan and the model view differ in style, in particular, with regard to line thicknesses and hatchings, lowering the similarity of the images and possibly impacting feature-based registration methods.
- Before the actual registration, the perspective of the escape plan image needs to be corrected or reduced to a minimum, which itself might require explicit input.
- The structural geometry of the escape plan might not fit the BIM model. These differences in as-is and as-planned impair the similarity beyond drawing style.

Therefore, the automatic registration of escape plan and model view image remains subject to further research.

Instead, we employ user interaction to obtain the correct transformation. The user selects four matching 2D point pairs across both the photograph and the model's plan view, denoted as $\{(x_1, y_1), \dots, (x_4, y_4)\}$ and $\{(x'_1, y'_1), \dots, (x'_4, y'_4)\}$, respectively. These points, expressed in homogeneous coordinates as $(x_1, y_1, 1), (x'_1, y'_1, 1)$, serve as the basis for determining the transformation matrix M , which is crucial for mapping the photograph onto the model's plan view. The transformation matrix M is derived by solving the linear equation system defined in Eq. (1).

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} = M \cdot \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} \quad (1)$$

In this equation system, each corresponding pair of points directly constrains the matrix entries m_{ij} . Through the collection of four pairs, a well-determined system of linear equations is formed. The solution to this system yields the matrix entries m_{ij} such that of M accurately maps the selected points in the photograph to their corresponding points in the model's plan view. For implementation details, the reader is

referred to OpenCV.² Finally, the photograph and all the associated keypoint vectors are transformed by applying M to determine the FSE components' locations in the model COS.

3.6. BIM model enrichment

Once the physical locations of FSE components and their respective types are computed, the next step is to enrich the building model with these components.

The manual insertion of FSE components into a 3D building model is typically performed in commercial Computer-aided Design (CAD) software. However, since the automatic insertion of FSE to the building model does not require a graphical interface, and the use of open source software ensures vendor neutrality, we choose to utilize the IfcOpenShell³ to edit the model's IFC file directly.

To ensure robust future information querying and retrieval, a systematic approach is followed during the enrichment of the IFC model with FSE equipment. This involves modeling components using the most appropriate IFC class and classifying their type using the most appropriate IFC predefined types as well as Uniclass codes.

In terms of the IFC schema,⁴ the components are added to the model's IFC file using the classes and predefined types listed in Table 2. Additionally, each component is attributed a Uniclass code for finer classification. For precise placement, an *IfcLocalPlacement* is created at the desired location for each component. The corresponding 3D representations of the components are pre-modeled and saved as OBJ-files. They are included into the BIM model using *IfcShapeRepresentation*, each associated with one *IfcLocalPlacement*. It is noted that positional adjustments, such as relocating inserted components to the closest wall or resolving clash detections, are beyond the scope of this study.

4. Experimental results

To validate the proposed method for processing escape plans, this section provides insights into the characteristics of the utilized datasets, presents test results from trained deep learning models, and demonstrates the pipeline's application in a representative case study building. Following the methodology described in Section 3, first, the acquisition of training and test datasets is detailed in Section 4.1. Subsequently, results are reported, specifically focusing on the validation of the ROI cropping model (Section 4.2), the keypoint detection model (Section 4.3), and the story extraction (Section 4.4). Finally, Section 4.5 showcases the enrichment of the exemplary campus building.

4.1. Datasets

In this study, a total of 332 escape plan images are used for training the deep learning models. First, 224 real escape plan images are collected in campus buildings of the Ruhr University Bochum (RUB). Second, to enrich the diversity and increase the size of the dataset, 108 additional escape plan images obtained from the web are fed to the synthetic training image generation pipeline (cf. Section 3.3.3). Finally, all training images are annotated with ROI and symbol keypoint annotations, to be used in both training scenarios.

An analysis of symbol occurrences per class, detailed in Table 1, highlights certain imbalances within the dataset. Notably, the campus dataset lacks instances of fire fighting equipment, emergency phone, and fire ladder symbols, which are, admittedly, quite rare in the synthetic dataset as well. The majority of symbols is represented by fire extinguishers, manual call points, and fire hose reels. However, there is

² <https://docs.opencv.org/3.4/index.html>

³ <http://ifcopenshell.org> (Accessed: 15 December 2023)

⁴ https://standards.buildingsmart.org/IFC/RELEASE/IFC4_3 (Accessed: 15 December 2023)

Table 2
Identified FSE components as IFC objects, additionally specified with Uniclass codes.

Component	IFC class	Uniclass
Fire extinguisher	IfcFireSuppressionTerminal.FIREHYDRANT	Pr_40_50_28_64 (Portable fire extinguishers)
Fire hose reel	IfcFireSuppressionTerminal.HOSEREEL	Pr_65_52_38_30 (Fire hose reels)
Manual call point	IfcAlarm.MANUALPULLBOX	Pr_75_80_30_50 (Manual call points)
Emergency phone	IfcAudioVisualAppliance.TELEPHONE	Pr_70_75_88_03 (Analogue telephone handsets)
Fire ladder	IfcStair.USERDEFINED	Pr_25_30_85_02 (Access ladders)

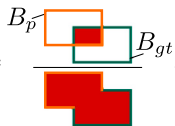
still a significant class imbalance among these three classes, suggesting a potential bias in the trained keypoint detection model, particularly favoring fire extinguishers and manual call points. It is noteworthy that each escape plan in the dataset features exactly one ROI.

To evaluate the trained models, a separate test set of real images is compiled from two RUB campus buildings, namely the *IC* and *ZESS* buildings, comprising a total of 53 escape plan images. Both deep learning models undergo testing on these plans, and the results are detailed in the following sections. Additionally, the results are showcased through a demonstration escape plan, serving as a case study subject for the entire BIM enrichment pipeline. The case study focuses on the *IC* building on the RUB campus. The escape plan of the facility's top floor is acquired as part of the test set and annotated, and a BIM model in IFC format is separately created for subsequent enrichment.

4.2. Floor plan region cropping

When assessing the ROI detection model, the key metric of concern is the completeness of the extracted floor plan view. While ROI detection is beneficial for enhancing keypoint detection performance and preventing false positives by excluding the legend, it introduces a potential risk of unintentionally cropping the floor plan view itself. This, in turn, may impede the functionality of registration methods or result in overlooked symbols during the keypoint detection phase. Therefore, the evaluation of the floor plan cropping step primarily focuses on metrics like the Recall R and the Intersection over Union (IoU), which gauge the model's ability to retain relevant information accurately. Additionally, the Average Precision (AP) is reported as well since it is a standard metric for object detection tasks like this.

The IoU measures the spatial overlap between the predicted ROI and the ground truth, defined as

$$IoU = \frac{\text{area of intersection}}{\text{area of union}} = \frac{|B_p \cap B_{gt}|}{|B_p \cup B_{gt}|}, \quad (2)$$


where the *union* refers to the combined area encompassed by both the predicted (B_p) and ground truth (B_{gt}) bounding boxes, while the *intersection* represents the overlapping region between them. This concept is pivotal in determining the fidelity of object localization, as a higher IoU indicates a greater spatial agreement between the predicted and actual objects. In this context, a high IoU is desirable to minimize overlooked symbols while maximizing the portion of the cropped image containing actual content. Also, the IoU is interpreted as a similarity measure for comparing ground truth objects with predicted objects. This property is used to determine whether a detection is a true or false positive. Typically, a threshold of 0.5 IoU is defined for that purpose [80]. However, in our case, a higher threshold of 0.85 is set to minimize overlooked symbols.

The recall value R assesses the proportion of relevant items (in this case, the floor plan details) successfully retrieved by the model, calculated as

$$R = \frac{\text{true positives}}{\text{all relevant objects}}, \quad (3)$$

while, in this case, there is only one relevant object per image, anyway. Therefore, $R = 1$ if the ROI is detected, otherwise, $R = 0$. Hence, if R is reported for the whole dataset, it directly reflects the proportion of

correctly processed test images, where a high R value indicates a good ROI detection performance. Finally, AP is a widely recognized metric in object detection, which serves as a holistic measure of the model's overall accuracy, accounting for both precision and recall at varying confidence thresholds [81]. Both values R and AP depend on the chosen IoU threshold, where higher threshold restrict the evaluation to very good matches. These metrics collectively provide a comprehensive evaluation framework for the ROI detection model's performance in extracting floor plan views.

When applied to the test set, the trained model achieves $R = 1.0$ even with a high IoU threshold of 0.85, and $AP_{0.85} = 0.995$ accordingly, demonstrating its effectiveness for the task. Visual inspection of all test images confirms that predictions capture the entire floor plan region without leaving excess space. An example ROI detection is shown in Fig. 9(a). Two candidate bounding boxes are inferred, where one encompasses the desired floor plan geometry (confidence of 0.91), and the other mistakes a facility overview map for the floor plan (confidence of 0.81). This example highlights the use of the model's confidence value for each detection, as only the detection with highest confidence is chosen for further processing, while the rest is disregarded. Therefore, after considering the confidence, the correct bounding box is obtained, including the entire floor plan geometry, and excluding unnecessary image regions.

4.3. Symbol detection

To comprehensively evaluate the FSE localization performance, two key metrics are considered. First, to assess the overall performance of the keypoint detection model, the AP is reported with a threshold defined by the Object Keypoint Similarity (OKS) score, which, for an object with N keypoints, is computed by

$$OKS = \frac{1}{N} \sum_{i=1}^N \exp\left(-\frac{d_i^2}{2s^2k_i^2}\right), \quad (4)$$

where d_i is the Euclidian distance between the predicted and ground-truth position of the i th keypoint, and s and k_i are statistical values accounting for the object scale and standard deviations in keypoint annotation, respectively. Similar to IoU in object detection, OKS measures the similarity between the predicted and ground truth symbol skeleton, accounting for spatial proximity and scale, as detailed in the COCO Benchmark [82]. In the context of FSE symbol detection, OKS is crucial for quantifying the accuracy of localization for individual symbols within the extracted floor plan. For the computation of AP , the default threshold of $OKS = 0.50$ is used. Since the AP is computed per class, the mean Average Precision (mAP) is defined as

$$mAP = \frac{1}{C} \sum_{c=1}^C AP_c, \quad (5)$$

where $C = 3$ is the number of classes in the test dataset. Regarding per-class performance, the keypoint detection model achieves AP scores of 0.88, 0.83, and 0.49 for fire extinguishers, manual call points, and fire hose reels, respectively. The differences in performance reflect the class imbalance in the dataset. However, the mAP value of 0.73 and the high performance scores of fire extinguishers and manual call points underscore the model's general effectiveness for the task.

Secondly, the deviation between the predicted and ground truth locations of FSE components is computed, expressed as a distance in

Table 3
Symbol detection results on case study escape plan.

Component	Detected	Error (pixels)	Error (metric)
Fire extinguisher 1	✓	0 px	0 cm
Fire extinguisher 2	✓	2 px	7 cm
Fire extinguisher 3	✓	3 px	11 cm
Fire extinguisher 4	✓	39 px	144 cm
Fire extinguisher 5	✓	3 px	11 cm
Fire extinguisher 6	✓	2 px	7 cm
Fire extinguisher 7	✓	2 px	7 cm
Fire extinguisher 8	✓	1 px	4 cm
Fire extinguisher 9	✓	2 px	7 cm
Fire extinguisher 10	✓	43 px	159 cm
Fire extinguisher 11	✓	9 px	33 cm
Fire extinguisher 12	✓	2 px	7 cm
Manual call point 1	✓	3 px	11 cm
Manual call point 2	✓	12 px	44 cm
Fire hose reel 1	✓	230 px	851 cm
Fire hose reel 2	✗	–	–

meters. This distance metric is essential for two reasons. Firstly, a low error in meters is meaningful in real-world scenarios, as it determines the applicability of the proposed method. Secondly, the pixel error is influenced by image resolution and camera hardware, whereas the error in meters provides a standardized measure, allowing for comparisons across different extraction pipelines or input file resolutions.

To showcase the accuracy of the pipeline, Table 3 provides a list of components on one floor, presenting the prediction results alongside the respective error distances in pixels and meters. The detections are visualized in Fig. 9(b). Each model output with a confidence score of over 0.5 is visualized by a rectangular tag in the input image, and indicates the inferred location on the drawing with an associated dot. First and foremost, it is evident that the model can detect the FSE symbols with the exception of one fire hose reel symbol, which may be a result of the class distribution imbalance discussed in Section 4.1. Second, most of the detected objects' positions are inferred with low positional errors, with the exceptions of fire extinguishers #4 and #10, as well as fire hose reel #1. Notably, all the fire extinguishers' locations are inferred to be in the correct room, and only the wall placements are incorrect, leading to positional errors of 1.44 m and 1.59 m. In these instances, admittedly, the precise location is quite ambiguous even for humans, which is considered an inevitable flaw in the drawing style of escape plans, ultimately limiting the reliability of the proposed method. Third, although the fire hose reel #1 is clearly indicated on the escape plan with an auxiliary line, it is completely misplaced by the detection model, as it is not even inferred to be in the correct room. Evidently, the model is not able to recognize the auxiliary line in this symbol instance. All in all, the model proves to be generally capable of the task, albeit with minor restrictions likely due to discussed issues with the dataset, and with limitations imposed by the nature of escape plans.

It is worth noting that the results in Table 3 should be interpreted carefully. On the one hand, most the reported positional errors amount to 11 cm or less, and, although technically correct, are practically insignificant when considering real-world scenarios. Whether the FSE components are placed exactly as shown on the plan or a bit off (by a few centimeters) is not important. The escape plan remains effective as long as the components are roughly in the same area, in particular in the same room. On the other hand, if errors amount to some meters, this could lead to FSE components being placed in neighboring rooms, possibly affecting maintenance operations and model analyses regarding fire safety.

Another thing to consider is that the error measures presented here do not account for any potential errors induced by the registration process. It is computed solely based on the escape plan and the known building story outline. However, if conducted and checked carefully (e.g. using the registration Mean Square Error) it is assumed that the registration process introduces negligible error, for reasons similar to those stated above.

4.4. Text extraction

The evaluation of the text detection and recognition models focuses on their predefined task of extracting building story numbers from floor plans. As these models are pre-trained and have demonstrated effectiveness in detecting text elements on floor plans, they are not quantitatively evaluated for all text elements on the drawing (cf. [74]).

However, to gauge the efficacy of the text extraction pipeline, a test is conducted on the provided test set. In this evaluation, the primary interest lies in the accurate extraction of building story numbers. Results indicate that in 49 out of the 53 images (92.5%), the text extraction pipeline successfully identifies and extracts the building story number. The story number could not be identified in the remaining 4 cases automatically, however, this is easily addressed by prompting the user to enter the missing floor number. This underscores the model's capability to fulfill its intended purpose within the given context.

4.5. BIM model enrichment

The enrichment of the BIM model with FSE components is a pivotal step in realizing a comprehensive safety inspection framework. Leveraging the IfcOpenShell toolkit for direct manipulation of the BIM model's IFC file, this study successfully enriches the BIM model with semantically-rich 3D representations of identified FSE components.

Given a preference for open-source software, the results are visualized in Blender, leveraging the BlenderBIM Add-on.⁵ The enriched model, showcased in Fig. 10, not only visualizes the spatial positions of safety components but also ensures their categorization in adherence to the IFC schema and Uniclass classifications. However, it is crucial to acknowledge that the exact positional alignment with the physical structures like walls may vary. For instance, the 3D representation of extinguishers may not accurately reflect their attachment to walls, i.e., components float in their proximity or overlap partially. Also, information about the attachment height of the FSE components cannot be deduced solely from a top-view drawing such as the escape plan. Therefore, it is necessary to make an assumption based on a standard height, which might vary in accordance with national regulations. While this introduces a degree of uncertainty, it is typically minimal and does not hinder the creation of meaningful information displays, given that it can generally be presumed that the components are within arms' reach of most adults. Beyond the immediate visual enhancement, this enrichment sets the stage for various use cases, including fire safety inspection.

5. Discussion

5.1. Implications

This study presents a novel method with the potential to streamline the enrichment of existing building models by incorporating FSE components. In particular, the method addresses crucial gaps identified in a systematic review by Schönfelder et al. [26]. The review underscored the limited exploration of the as-is modeling of (and enrichment of BIM models with) small building components, focusing primarily on information extraction from building documentation. This study contributes to fill this gap by integrating secondary building components like FSE directly into existing IFC files, showcasing a holistic approach to retrospective building modeling.

The innovation of the study is two-fold. First, it introduces a semi-automatic BIM model enrichment method that relies on escape plans as the primary data source. Notably, the method minimizes the need for user input, with manual intervention only required during the registration step of the process. Second, the study surpasses previous

⁵ <https://blenderbim.org> (Accessed: 15 December 2023)

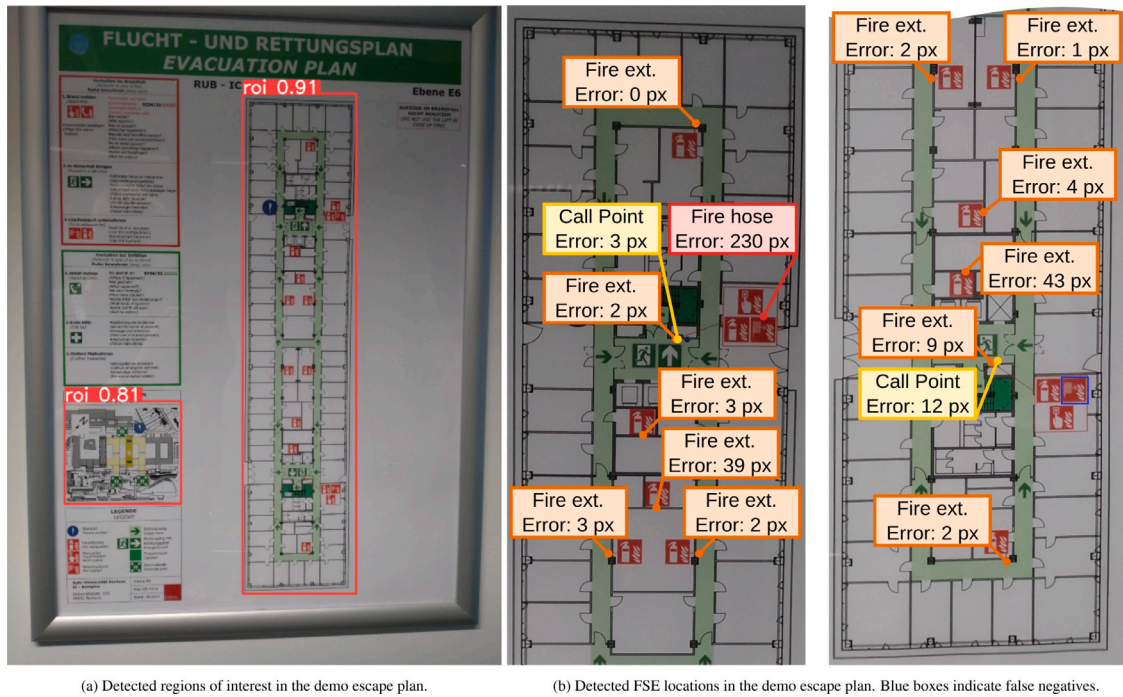


Fig. 9. Visualized localization results in the demo escape plan.

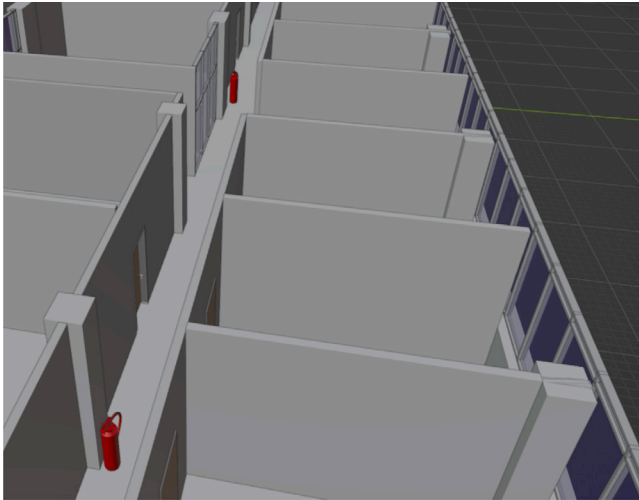


Fig. 10. Visualizing the enriched building model in BlenderBIM.

escape plan digitization approaches by introducing several novel developments. These include the extraction the ROI from escape plans (as described in Section 3.2), the adoption of keypoint-based symbol detection as opposed to bounding box-based detection (as described in Section 3.3), and the implementation of OCR for story number extraction (as described in Section 3.4). These advancements underscore the strengths and originality of the proposed method for BIM model enrichment.

The case study highlights the applicability of keypoint detection models beyond their traditional application in human pose estimation tasks, showcasing their relevance in the context of escape plans. Although the model trained in this study exhibits a lower performance

score compared to Mask R-CNN's application in pose estimation benchmark datasets ($AP = 89.2\%$ [72],⁶) potentially due to the constraints of a smaller training set, it lays the groundwork for possible expansions beyond the recognition of FSE symbols. While this study aims to enrich BIM models, and, therefore, focuses on FSE equipment symbols, escape plans exhibit more symbols worth extracting. For example, green safety symbols and directional arrows could be recognized to extract first aid equipment and escape route information automatically. Table 4 shows additional symbols to be found in escape plans with their respective potential benefits when extracted.

While this study is mainly motivated by integrating FSE components into an existing building model (Scan-to-BIM), it might inspire Scan-vs-BIM tasks as well. For example, using the escape plan analysis and comparing the results to a BIM model already populated with FSE information allows for inventory checks.

5.2. Shortcomings

While the proposed method excels at inferring the precise positions of fire safety equipment (FSE) components, it carries inherent limitations that warrant consideration.

Notably, the proposed method focuses on inferring the precise positions of the FSE components. While other aspects, such as the semantic data of these components, are not directly addressed, they hold significance for comprehensive inspections. For instance, specifications of fire extinguishers could help fire regulation compliance checking. However, by concentrating on accurately determining the component positions, this method establishes a solid foundation upon which future research can build to incorporate semantic information. Further exploration in this direction could lead to a more holistic approach, combining positional inference with semantic data analysis for even more insightful digital models. In any case, given that learning-based methods for extracting semantic information from escape plans cannot

⁶ <https://paperswithcode.com/sota/keypoint-detection-on-coco-test-challenge> (Accessed: 20 February 2024)

Table 4
Other symbols in escape plans with potential applications.

Symbol	Name	Potential application
	First aid kit	Model enrichment for safety inspection
	Defibrillator	Model enrichment for safety inspection
	Escape ladder	Escape route extraction
	Exit	Escape route extraction
	Exit	Escape route extraction
	Assembly point	Escape route extraction

ensure absolute accuracy, it is advisable to enhance the BIM model not only with the specific FSE components, but also with corresponding image crops in which they were identified, or other relevant raw data for visual verification.

Another limitation arises from the assumption that the analyzed escape plan accurately represents the locations of FSE components. Deviations between as-designed and as-is locations could potentially impact the utility of the enriched 3D model for visualization. Also, in general, it is noted that these approaches rely on the escape plans' correctness, bearing potential legal implications to the building owners if they turn out to exhibit false information. This emphasizes the importance of accurate and up-to-date escape plans in the proposed methodology.

While most steps in the proposed pipeline operate automatically, the transformation from photograph to model view remains computed based on user input. In this regard, algorithms like Scale-Invariant Feature Transform (SIFT) and Hough transform are potential tools to extract points or lines across the escape plan and model plan views that can then be matched to compute the required transformation. Also, leveraging deep learning-based segmentation methods or alternative registration approaches are worth exploring. However, it is noted that differences in the two images' style, resolution, lighting, and perspective might raise non-negligible challenges that need to be accounted for in any automated registration approach. Also, automated registration methods could encounter difficulties in large indoor spaces with a high degree of repetitiveness and, thus, could lead to ambiguous results. In future studies, such issues might involve user interaction to select one of multiple suggested alignments, if the registration is ambiguous.

5.3. Future research endeavors

The proposed methodology lays the groundwork for several promising avenues of future research, extending its capabilities and exploring broader applications:

First, while the current approach extracts positional information solely from escape plans, future research could leverage fire protection concept documents to infuse additional semantic details into the BIM model. This would provide a more comprehensive understanding of fire safety equipment components beyond their spatial locations.

Second, examining the campus dataset reveals a natural imbalance in class support, primarily caused by real-world building conditions (cf. Table 1). This imbalance may lead to false negatives in learning-based

symbol detection methods, as observed in Fig. 9(b). Future studies focusing on escape plans could address this by incorporating modifications to the deep learning model's loss function or implementing advanced data sampling techniques. Notably, deliberate sampling of training escape plans may not fully rectify this imbalance, given its reflection within each escape plan, where certain symbols are more prevalent than others.

Third, automating the registration process between escape plans and the IFC building model remains an open challenge. Subsequent research efforts should concentrate on this aspect of the pipeline, exploring techniques such as deep learning-based segmentation methods to extract floor outlines. Additionally, identifying key building components like doors could serve as anchor points for registration, enhancing the automation of this critical step.

Fourth, future studies could delve into the extraction of information concerning fire safety equipment that is not explicitly outlined in escape plans, with the aim of integrating these components into BIM models. Key elements such as sprinkler systems, fire doors, and smoke evacuators undoubtedly play a role in enhancing fire safety by extending available safe evacuation times. Therefore, their inclusion in BIM models holds significance for accurate emergency response simulations and strategies [83].

Finally, a novel suggestion for future exploration involves an interdisciplinary study where the proposed method is employed to create an inventory and derive a checklist for equipment checking. Subsequently, the building model and checklist could be utilized for inspection route planning, potentially leading to an autonomous inspection process executed by robots. Additionally, the inspection results could be used to further enrich the BIM model with semantic information about the FSE, e.g., the fire extinguishers' expiry dates, manufacturers, and types (CO₂, foam). Such an integrated approach holds the promise of showcasing the full spectrum of automation potentials in safety inspections, providing a robust framework for efficient and accurate assessments.

6. Conclusions

In conclusion, this study presents a novel methodology for enriching existing BIM models with FSE components extracted from escape plans. Leveraging advanced deep learning techniques, the proposed approach demonstrates promising outcomes in the automation of this process, offering several insights and avenues for further research.

The proposed method has potential to facilitate the enrichment of BIM models with FSE components, addressing widely unexplored aspects of retrospective building modeling, e.g., small building components modeling. Despite its achievements, the proposed methodology has limitations that should be considered. First, it primarily focuses on inferring precise positions of FSE components, neglecting semantic data aspects that could enhance comprehensive inspections. Second, the reliance on escape plans assumes their accuracy in representing as-designed and as-is locations, potentially impacting the usefulness of the enriched 3D model for visualization. Additionally, the manual user input required for calculating the coordinate transformation from photograph to model view remains a notable aspect that demands further exploration for automation.

Looking ahead, future research endeavors should aim to extract semantic information from fire protection concept documents, address class imbalances in symbol detection, automate the registration process, and explore an integrated approach for creating inventories, checklists, and autonomous inspection routes using the proposed method. These extensions could unlock additional layers of insight and functionality, advancing the potential for automated safety inspections in complex building environments, and ultimately enhance fire safety overall.

CRediT authorship contribution statement

Phillip Schönfelder: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Angelina Aziz:** Writing – review & editing, Writing – original draft. **Frédéric Bosché:** Writing – review & editing, Supervision. **Markus König:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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